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Machine Learning CS 4641-7641

Support Vector Machine

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Slides are inspired from Mahdi Roozbahani, Andrew Zisserman, Yaser S. Abu-Mostafa.

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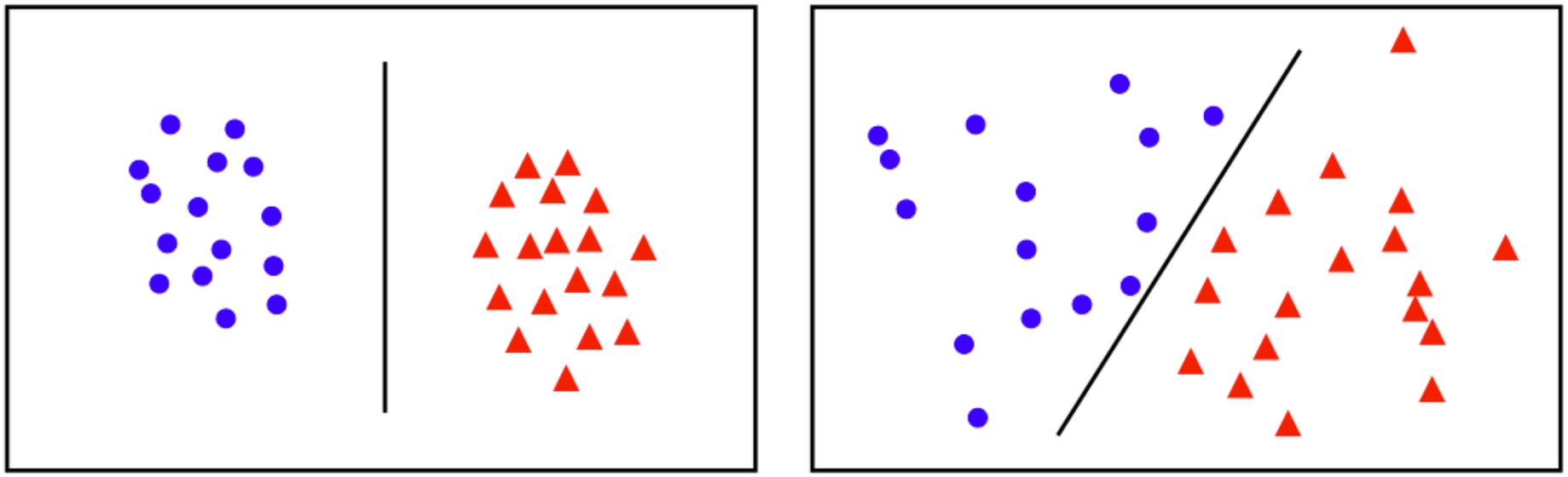
Outline

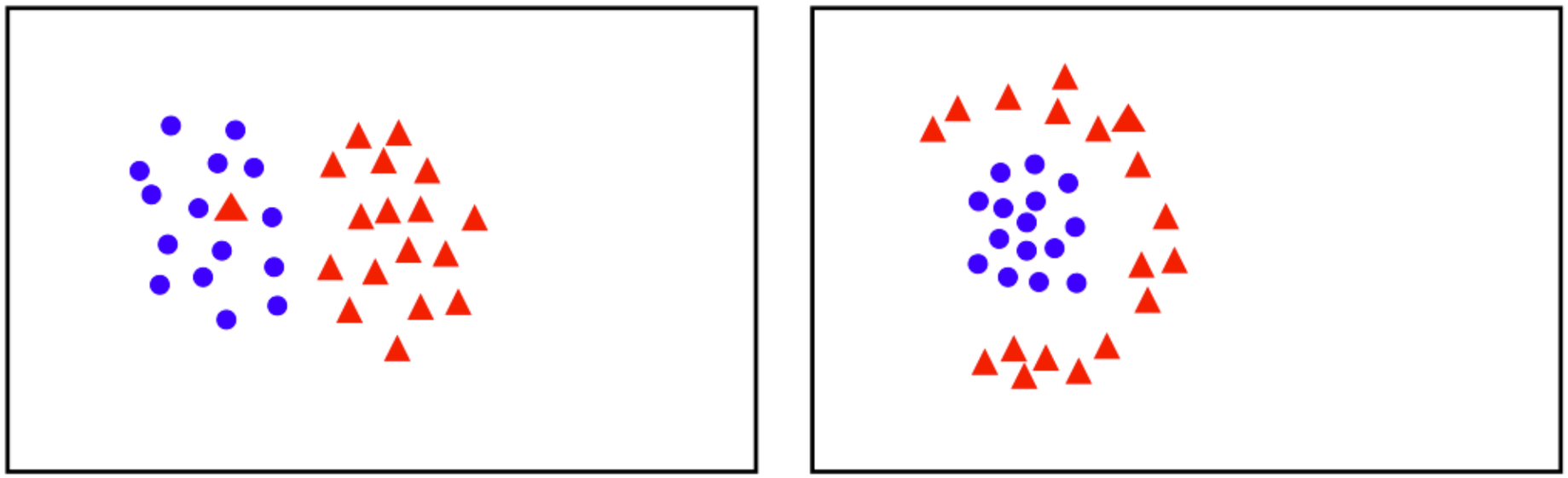
- Precursor: Linear Classifier and Perceptron
- Support Vector Machine
- Parameter Learning

Linear Separability

linearly separable



not linearly separable



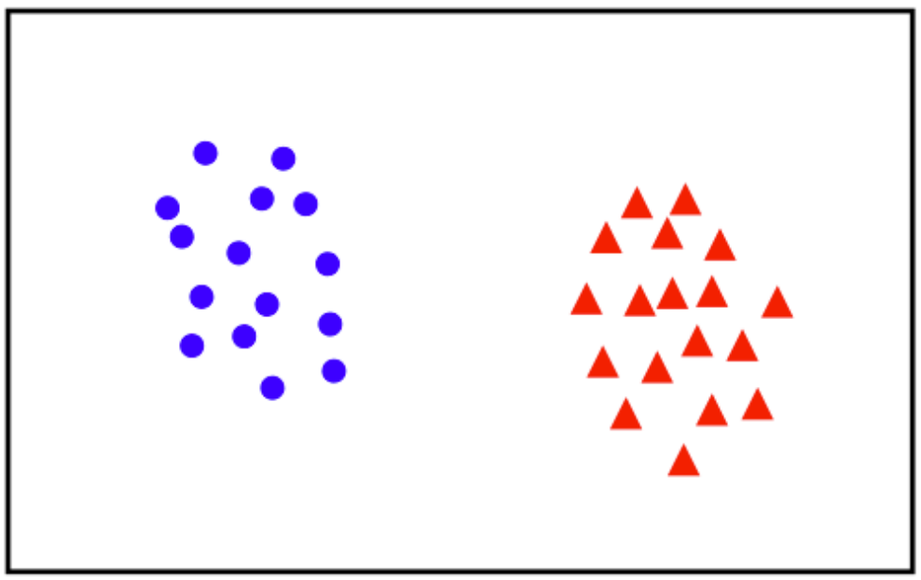
Binary Classification

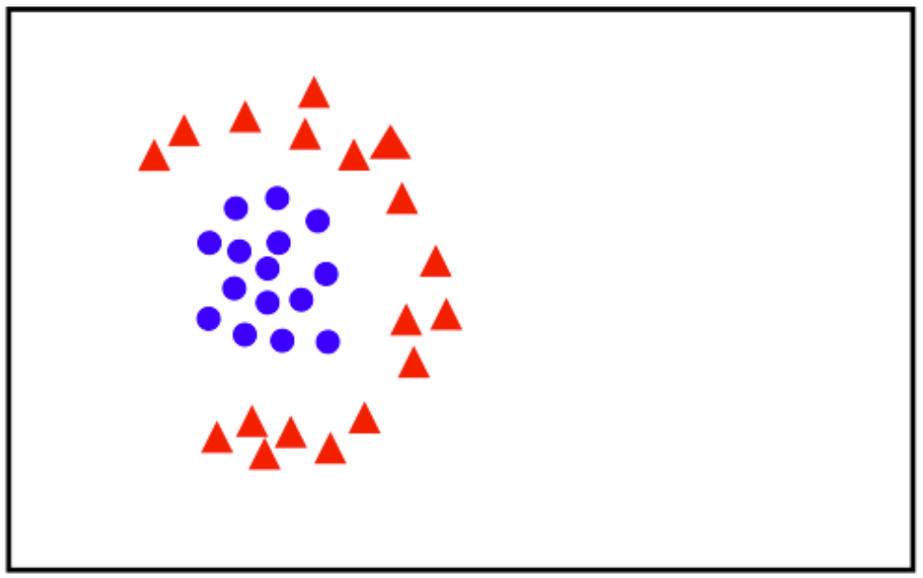
Given training data $(x^{\{i\}}, y^{\{i\}})$ for $i = 1 \ldots N$, with $x^{\{i\}} \in \mathbb{R}^d$ and $y_i \in \{-1, 1\}$, learn a classifier $f(\mathbf{x})$ such that

$$f(x^{\{i\}}) \begin{cases} \geq 0 & +1 \\ < 0 & -1 \end{cases}$$

+ + - -

i.e. $y^{\{i\}} f(x^{\{i\}}) > 0$ for a correct classification.

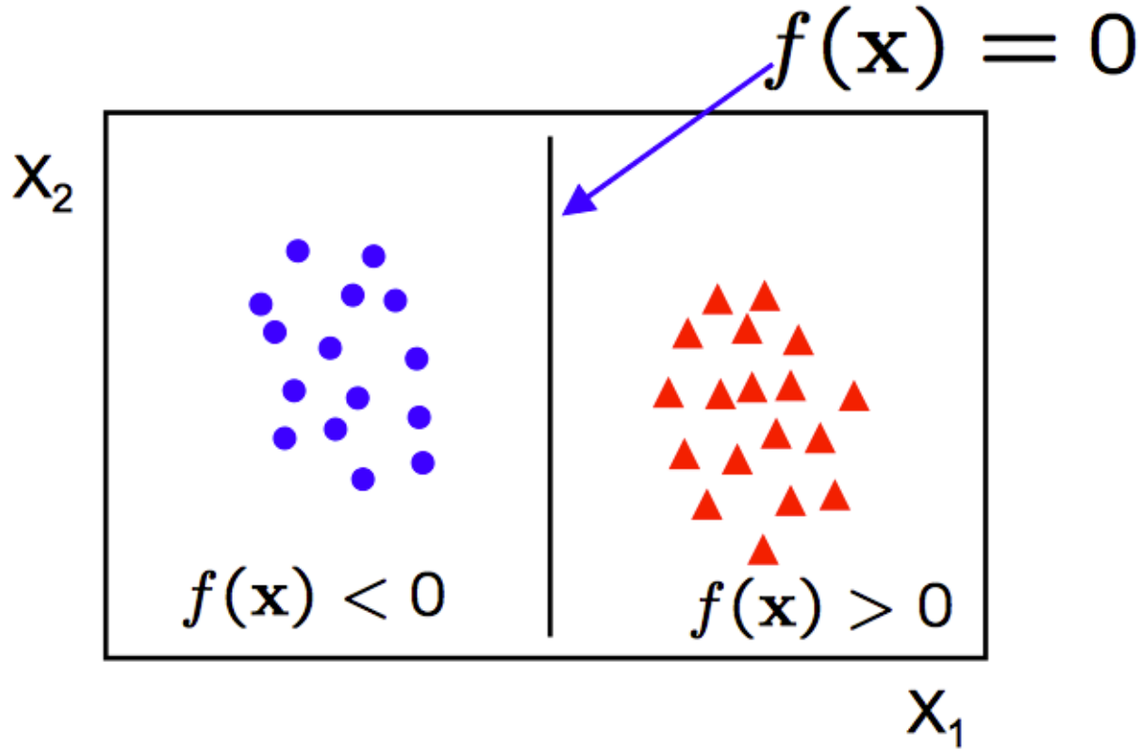




Linear Classifier

A linear classifier has the form

$$f(x) = x\theta + \theta_0$$

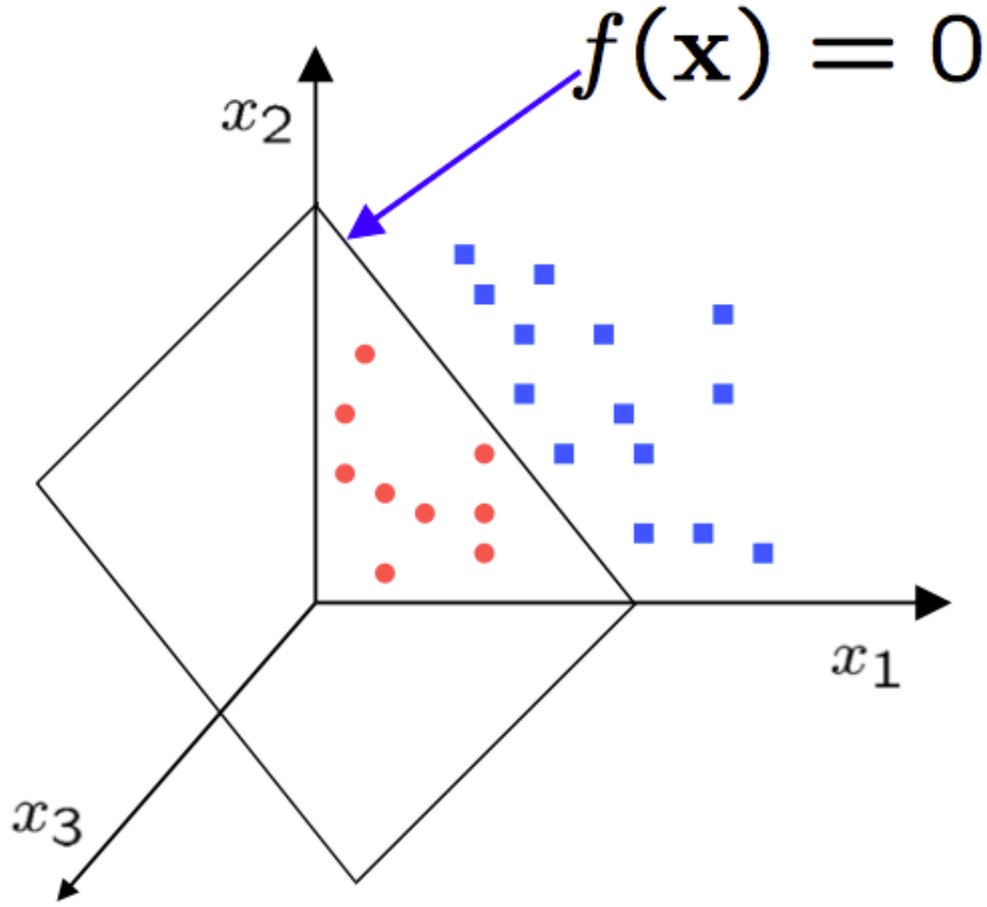


- in 2D the discriminant is a line
- θ is the normal to the line, and θ_0 the bias
- θ is known as the weight vector

Linear Classifier (higher dimension)

A linear classifier has the form

$$f(x) = x\theta + \theta_0$$



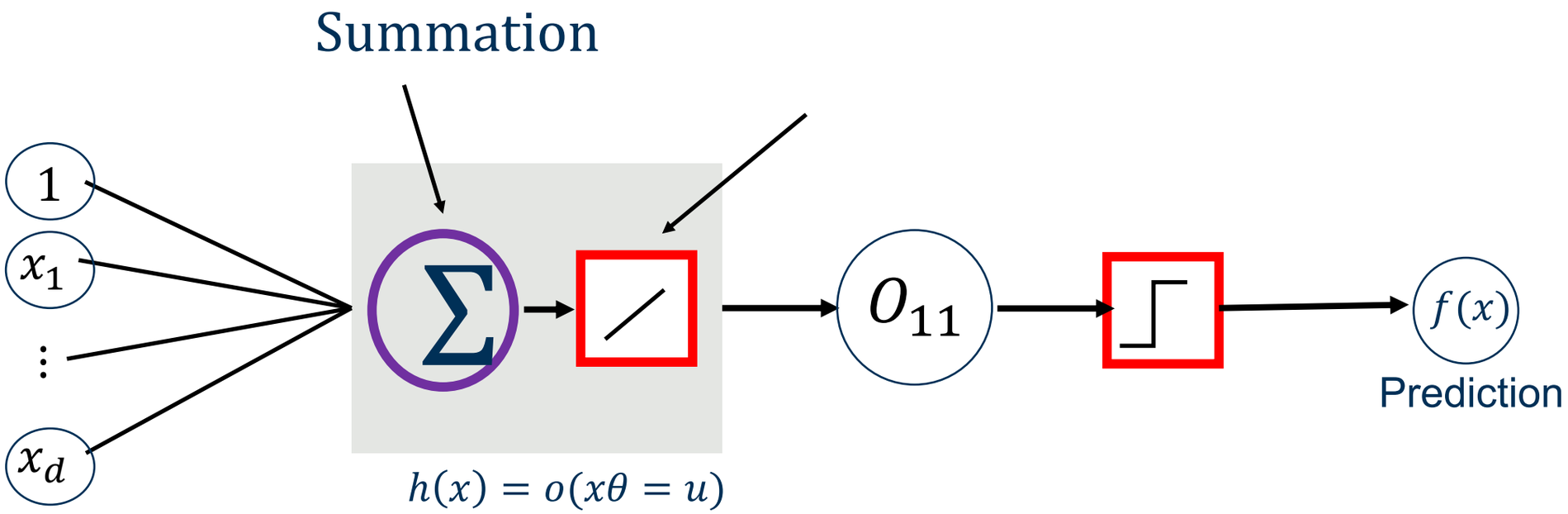
- in 3D the discriminant is a plane, and in nD it is a hyperplane

The Perceptron Classifier

Considering $\boldsymbol{x}$ is linearly separable and $\boldsymbol{y}$ has two labels of $\{-1,1\}$

$$f(x^{\{i\}}) = x^{\{i\}}\theta$$ Bias is inside θ now

How can we separate datapoints with label 1 from datapoints with label -1 using a line?



The Perceptron Classifier

Misclassified

Ex. $y^{\{i\}} f(x^{\{i\}}) < 0$

actual predicted

Perceptron Algorithm:

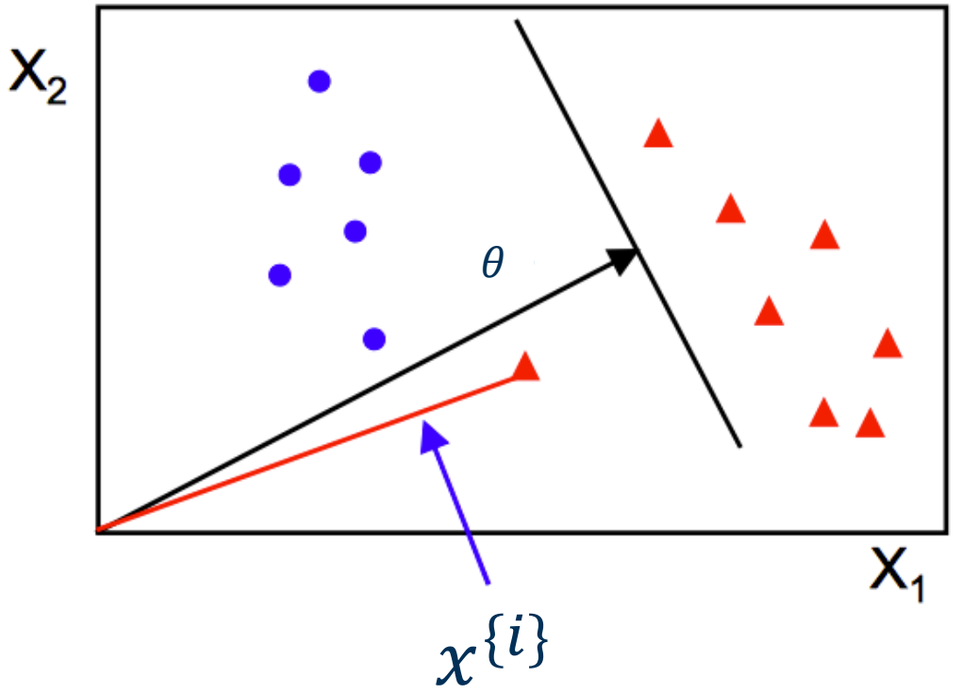
- Initialize θ
- Go through each datapoint $\{x^{\{i\}}, y^{\{i\}}\}$
- If $x^{\{i\}}$ is misclassified then $\theta^{t+1} \leftarrow \theta^t + \alpha y^{\{i\}} x^{\{i\}}$
- Until all datapoints are correctly classified

- Initialize θ
- Go through each datapoint $\{x^{\{i\}}, y^{\{i\}}\}$

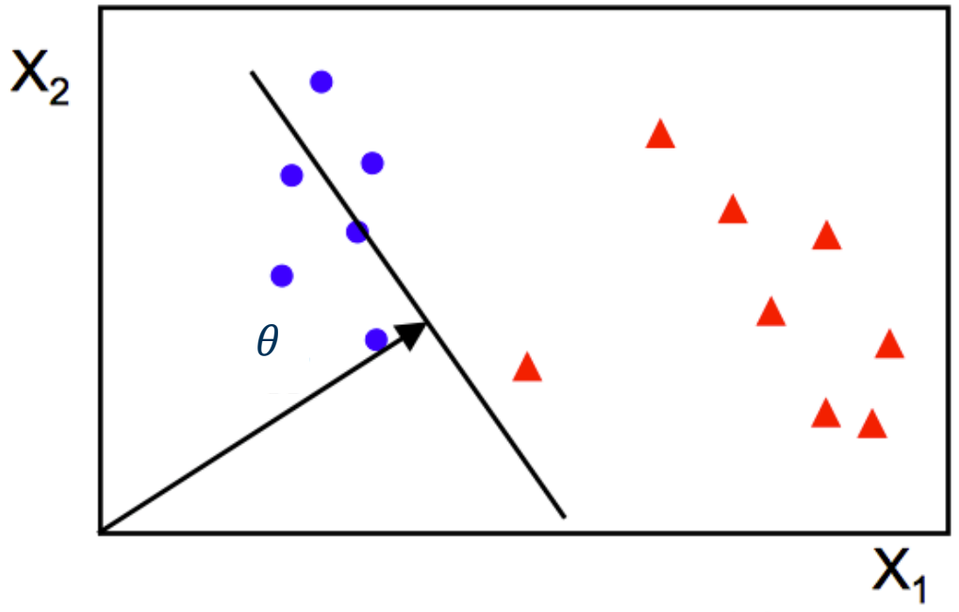
- If $x^{\{i\}}$ is misclassified then $\theta^{t+1} \leftarrow \theta^{t} + \alpha y^{\{i\}} x^{\{i\}}$

- Until all datapoints are correctly classified

before update



after update



$\theta^{t+1} \leftarrow \theta^{t} - \alpha x^{\{i\}}$

Recap

- Perceptron Algo

Perceptron update

If $x^{\{i\}}$ is misclassified then $\theta^{t+1} \leftarrow \theta^{t} + \alpha y^{\{i\}} x^{\{i\}}$

Misclassified point pulls θ in the direction that will flip the sign of $\theta \cdot x_i$ next time.

1. Multiply by y_i :choose the correct direction

- $y_i = +1$:move θ **toward** x_i – increase $\theta \cdot x_i$)
- $y_i = -1$:move θ **away** from x_i – decrease $\theta \cdot x_i$)

This flips the sign automatically so the dot product moves in the correct direction.

2. Multiply by x_i :fix the boundary using the misclassified point

- x_i points directly to *where* the mistake occurred
- Adding $+\alpha x_i$ rotates the decision boundary toward that point
- Adding $-\alpha x_i$ rotates it away from that point

The model adjusts based on the geometry of that specific data point.

3. Multiply by α: control the step size

- Small α → gentle correction. Large α → aggressive boundary movement

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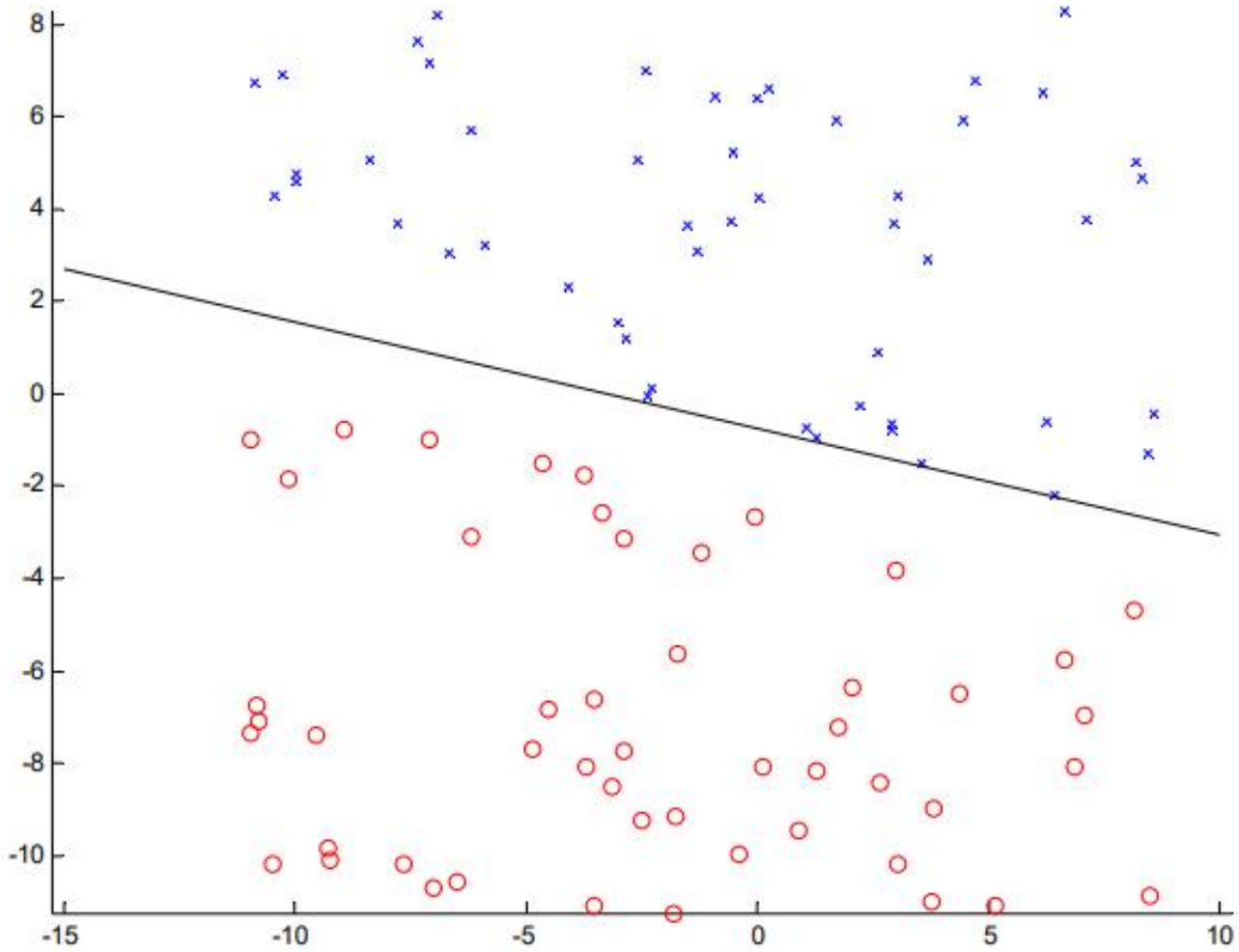
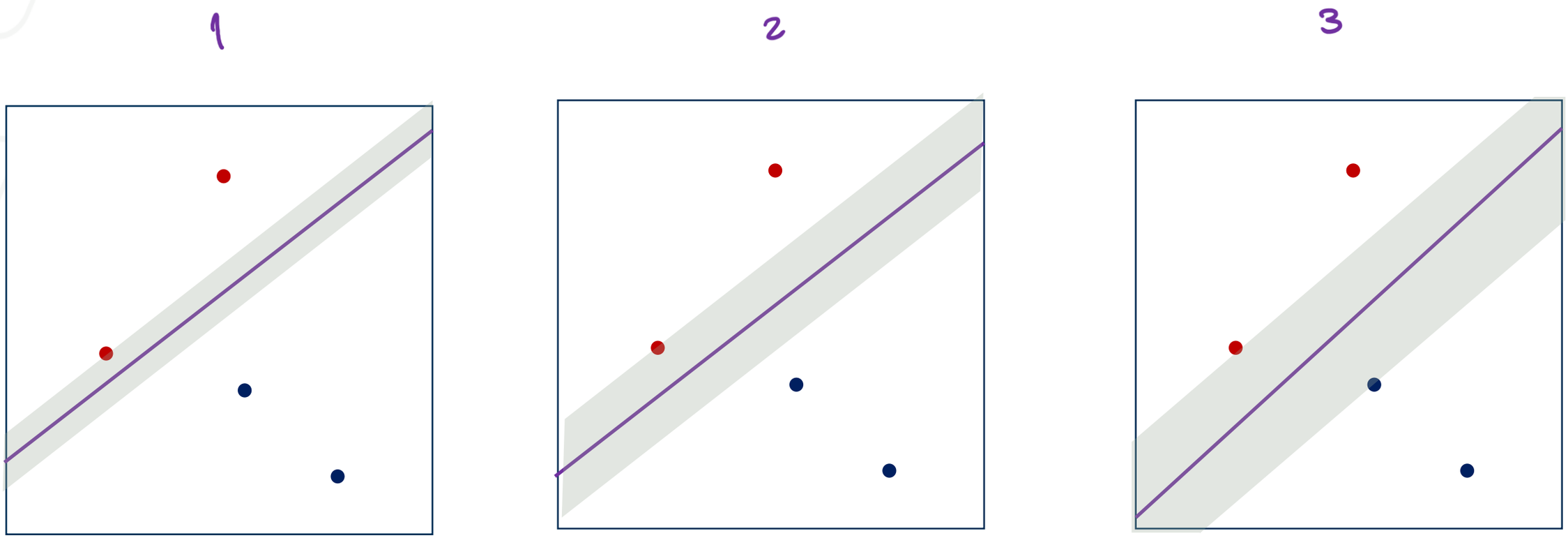
Perceptron example



- if the data is linearly separable, then the algorithm will converge
- convergence can be slow …
- separating line close to training data
- we would prefer a larger margin for generalization (better generalization)

Linear separation

We can have different separating lines



Which line is the best?

- Why is the bigger margin better?
- What θ maximizes the margin?

All cases, error is zero and they are linear, so they are all good for generalization.

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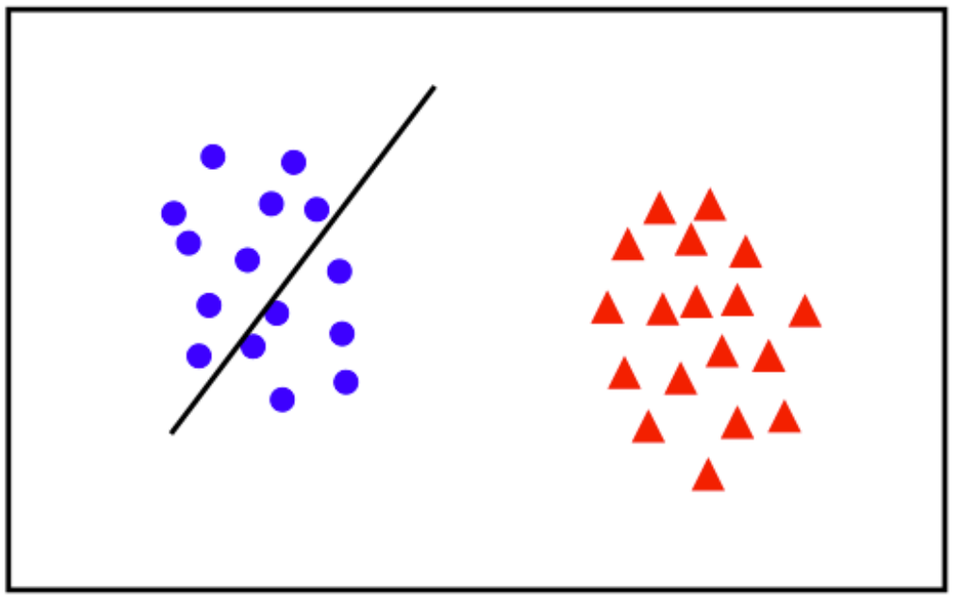
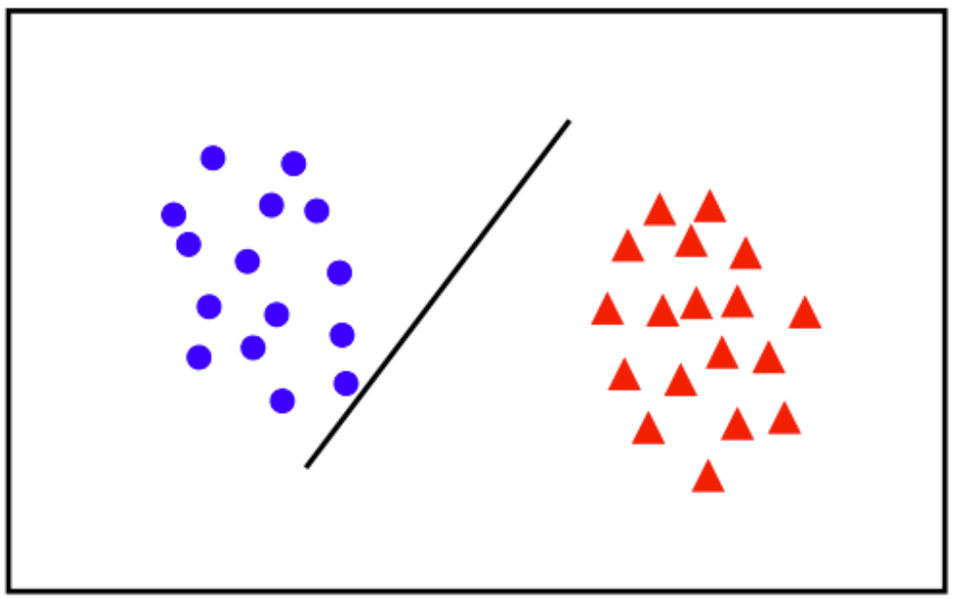
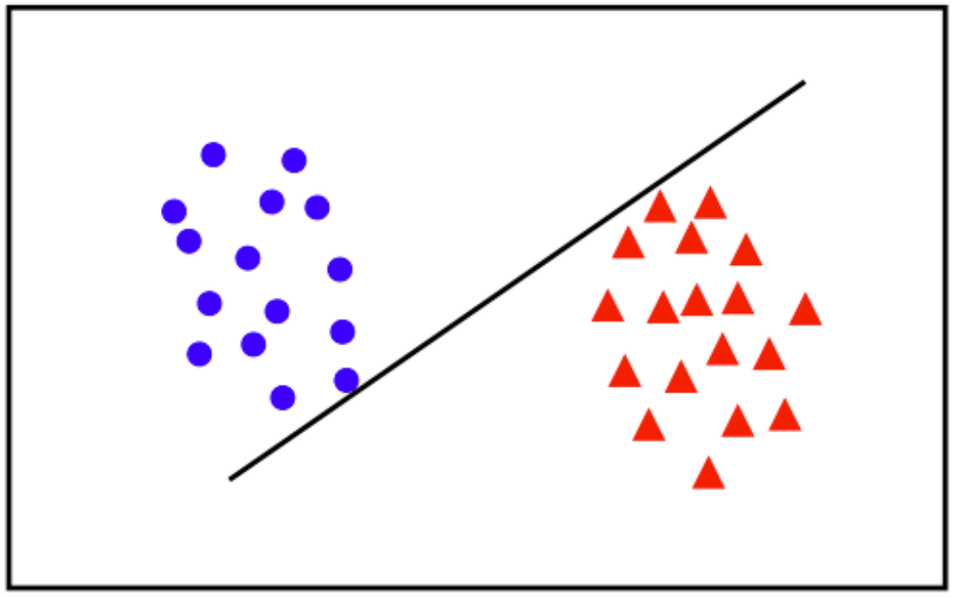
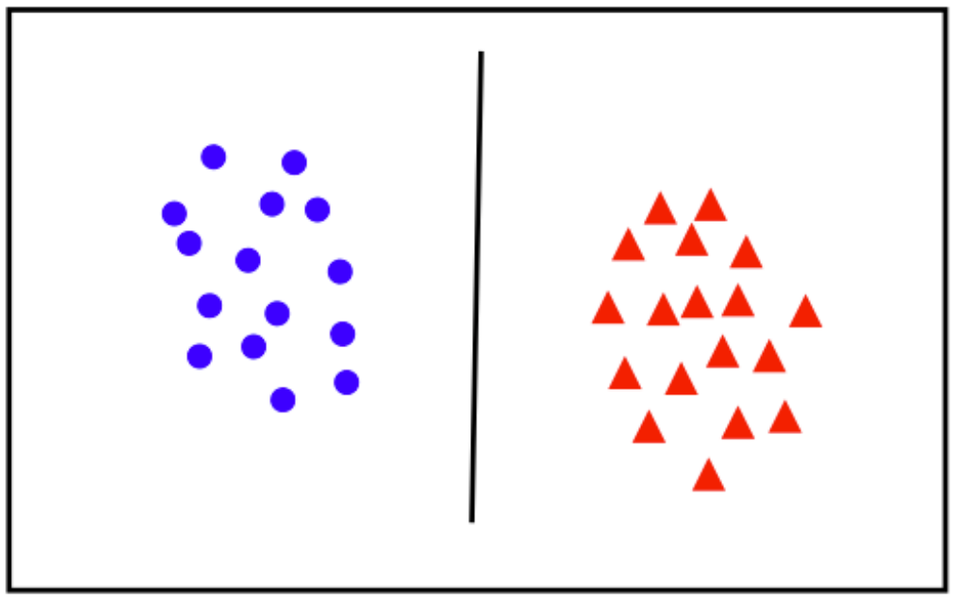
What is the Best θ?







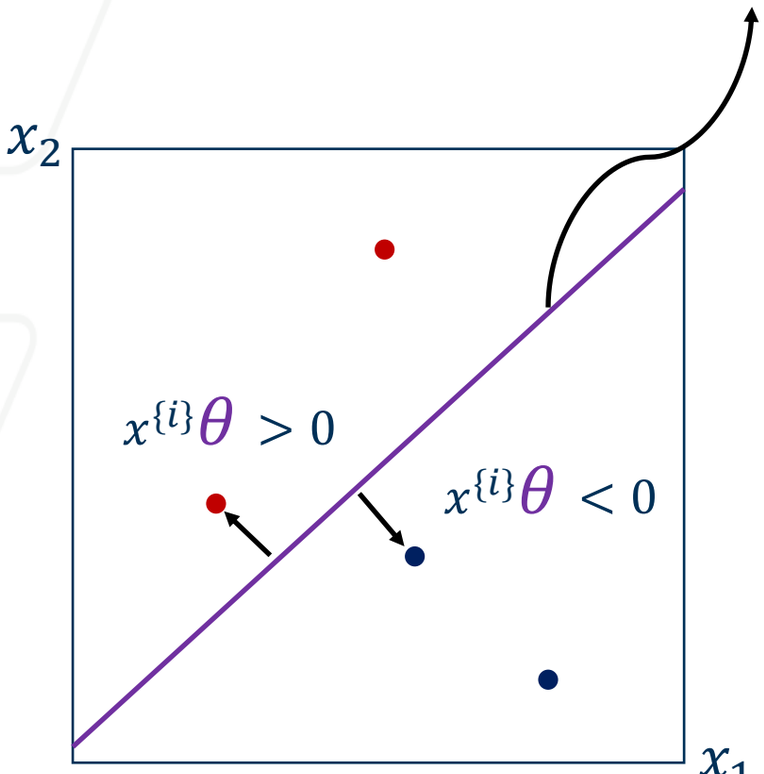


Outline

- Precursor: Linear Classifier and Perceptron
- Support Vector Machine ⬅
- Parameter Learning ⬅

Finding θ with a fat margin

Solution (decision boundary) of the line: $x\theta = 0$

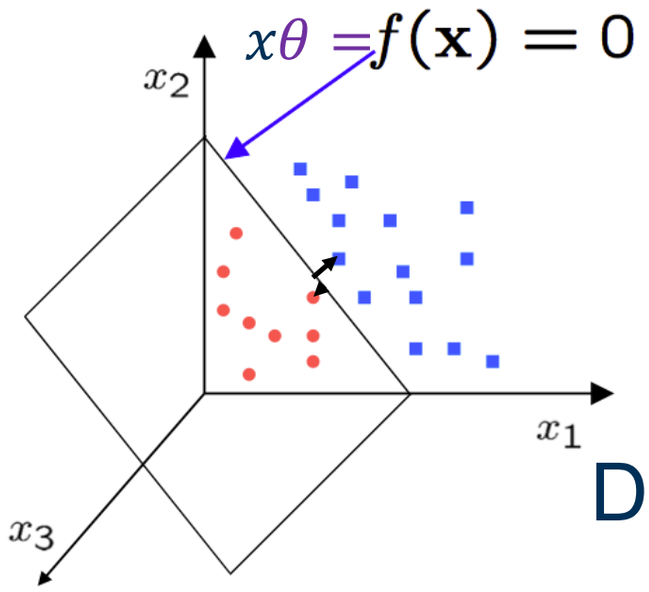


Let $\boldsymbol{x^{\{i\}}}$ to be the nearest data point to the line (plane):

$$\left|x^{\{i\}}\theta\right| > 0$$

Our line solution is $x\theta = 0$

Does it matter if I scale up or down θ for the decision boundary?



$$\left|x^{\{i\}}\theta\right| = 1 \rightarrow \text{normalization}$$

Let's pull out θ_0 from $\theta = (\theta_1, \ldots, \theta_d)$ and call it be b

Decision boundary would be: $x\theta + b = 0$

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Computing the distance

The distance between $\boldsymbol{x^{\{i\}}}$ and the line $x\theta + b = 0$ where $\left|x^{\{i\}}\theta + b\right| = 1$

The vector θ is perpendicular to the decision line.

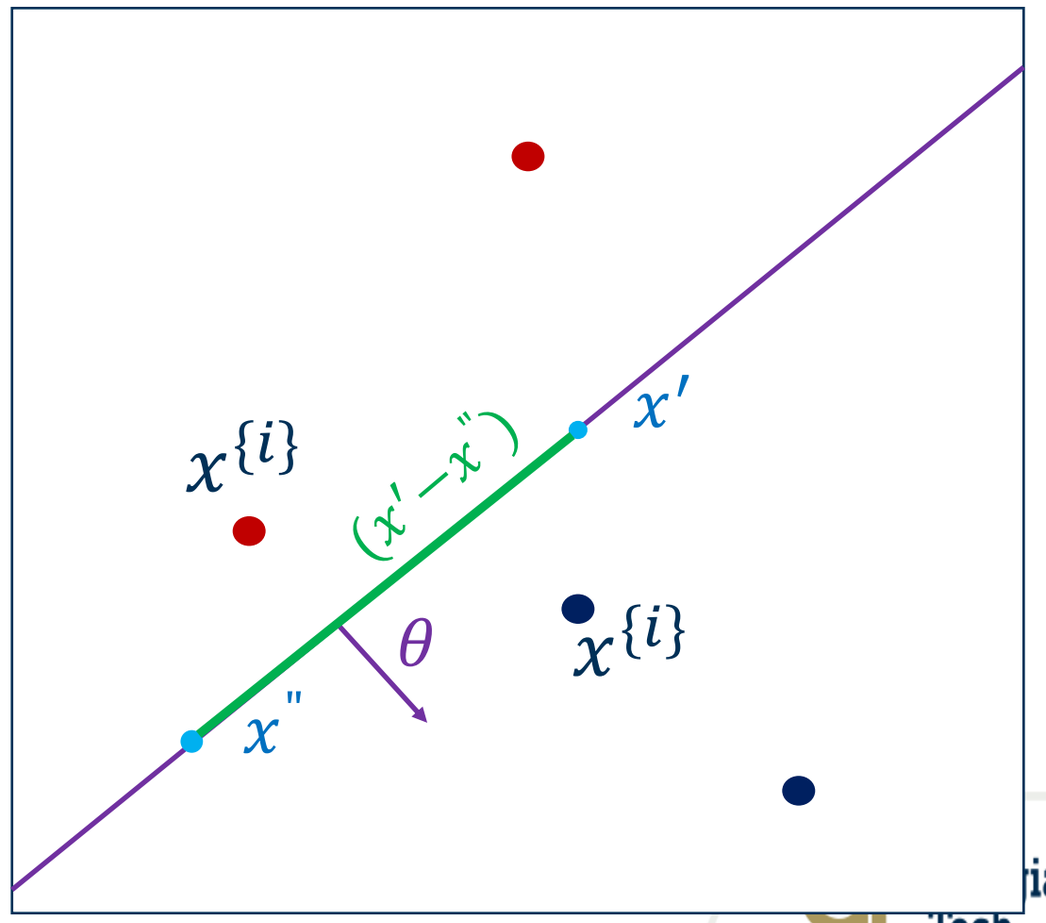
Consider x' and $x^{"}$ on the plane

$$x'\theta + b = 0 \quad \text{and} \quad x^{"}\theta + b = 0$$

$$\Downarrow$$

$$x'\theta + b = x^{"}\theta + b$$

$$\Rightarrow \quad (x' - x^{"})\theta = 0$$



What is the distance of my fat margin?

What is the distance between $x^{\{i\}}$ and the plane?

Let's take any point x on the line:

Distance would be projection of $(x^{\{i\}} - x)$ vector on θ.

To project the vector, we need to normalize θ to get the unit vector.

$\hat{\theta} = \frac{\theta}{||\theta||} \Rightarrow \text{distance} = \left| (x^{\{i\}} - x)\hat{\theta} \right|$ which is the dot product

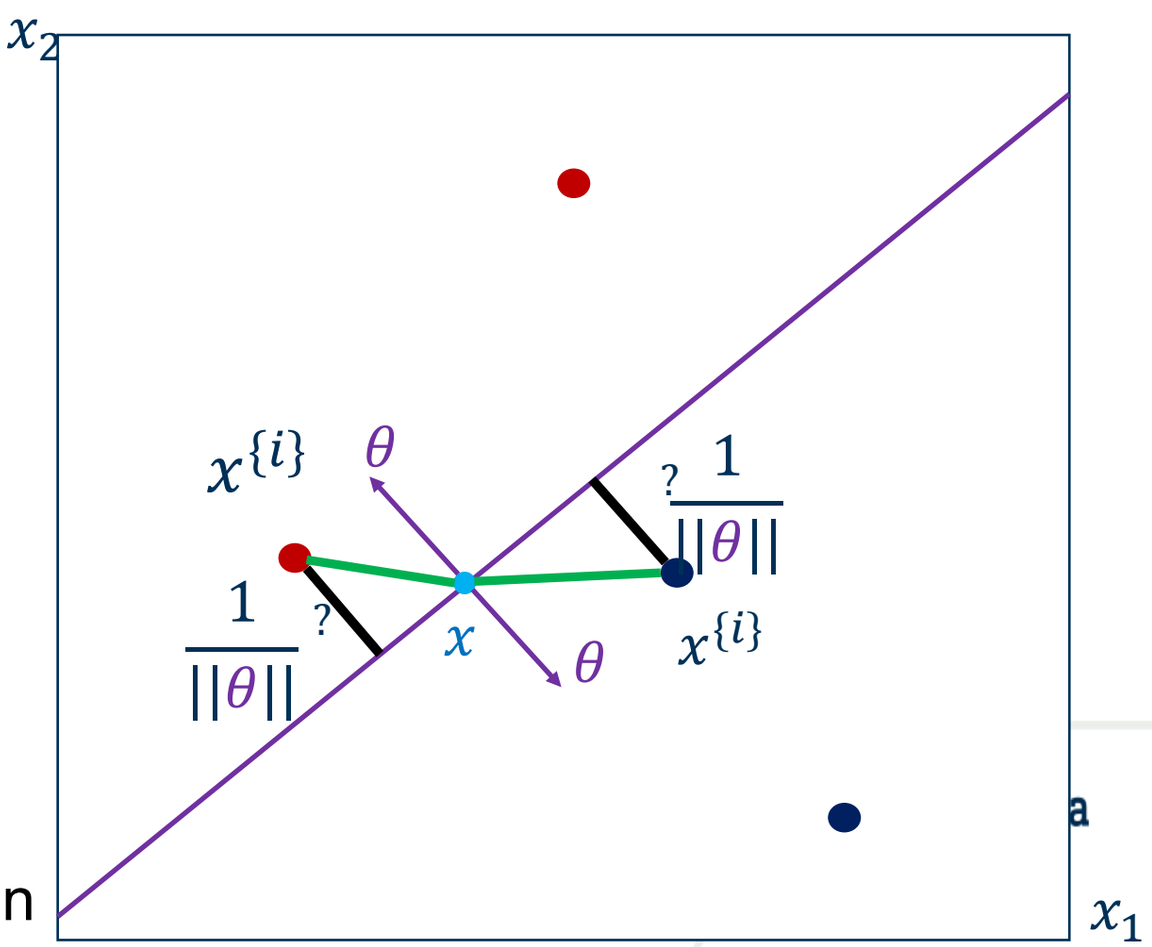
$$\text{distance} = \frac{1}{||\theta||}\left|(x^{\{i\}}\theta - x\theta)\right|$$

$$= \frac{1}{||\theta||}\left|(\underbrace{x^{\{i\}}\theta + b}_{\text{My constraint}} - \underbrace{x\theta - b}_{\text{A point on the decision line}})\right| = \frac{1}{||\theta||}$$

My constraint: $\left|x^{\{i\}}\theta + b\right| = 1$

A point on the decision line: $x\theta + b = 0$

The margin



Now we need to maximize the margin

Maximize $\frac{1}{||\theta||}$

Subject to Min value of $\left|x^{\{i\}}\theta + b\right| = 1 \Rightarrow nearest\ neighbour$

$i = 1,2, \dots, N$

There is a “min” in our constraining; it can be hard to optimize this problem(non-convex form)

Can I write the following term to get rid of absolute value?

$\left|x^{\{i\}}\theta + b\right| = y^{\{i\}}\left(x^{\{i\}}\theta + b\right) \Rightarrow$ for a correct classification

If min $\left|x^{\{i\}}\theta + b\right| = 1 \Rightarrow so\ it\ can\ be\ at\ least\ 1$

Maximize $\frac{1}{||\theta||}$

Subject to $y^{\{i\}}\left(x^{\{i\}}\theta + b\right) \geq 1$ for $i = 1,2, \dots, N$

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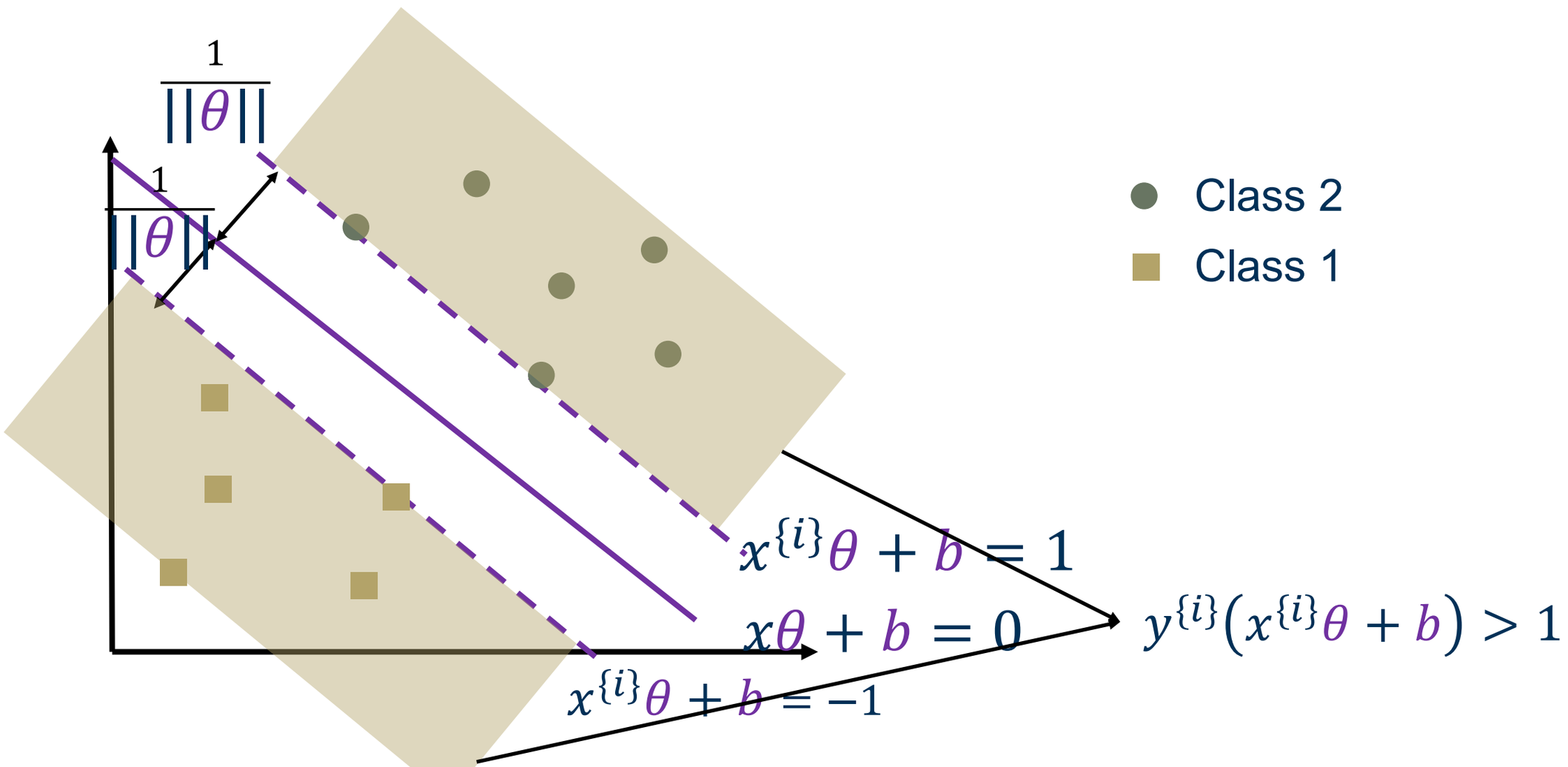

Maximize $\frac{2}{||\theta||}$

If $\theta \neq 0$, there exists a max value

Subject to $y^{\{i\}}\left(x^{\{i\}}\theta + b\right) \geq 1$ for $i = 1,2,\ldots,N$

Minimize $\frac{1}{2}\theta\theta^T$

Subject to $y^{\{i\}}\left(x^{\{i\}}\theta + b\right) \geq 1$ for $i = 1,2,\ldots,N$

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Constrained optimization

Minimize $\frac{1}{2}\theta\theta^T$

Subject to $y^{\{i\}}\left(x^{\{i\}}\theta + b\right) \geq 1$ for $i = 1,2,\dots,N$

$\theta \in \mathbb{R}^d, b \in \mathbb{R}$

Using Lagrange method:

But wait, there is an **inequality** in our constraints

We use **Karush-Kuhn-Tucker (KKT)** condition to deal with this problem

$g(x) = y^{\{i\}}\left(x^{\{i\}}\theta + b\right) - 1$ $\alpha = lagrange\ multiplier$

We need to optimize
$\theta, b, and\ \alpha$

1) Stationary

2) $g(x) \geq 0$ Primal feasibility

3) $\alpha \geq 0$ Dual feasibility

4) $g(x)\alpha = 0$ Complementary slackness

$$\Rightarrow \begin{cases} g(x) > 0, & \alpha = 0 \\ \alpha > 0, & g(x) = 0 \end{cases}$$

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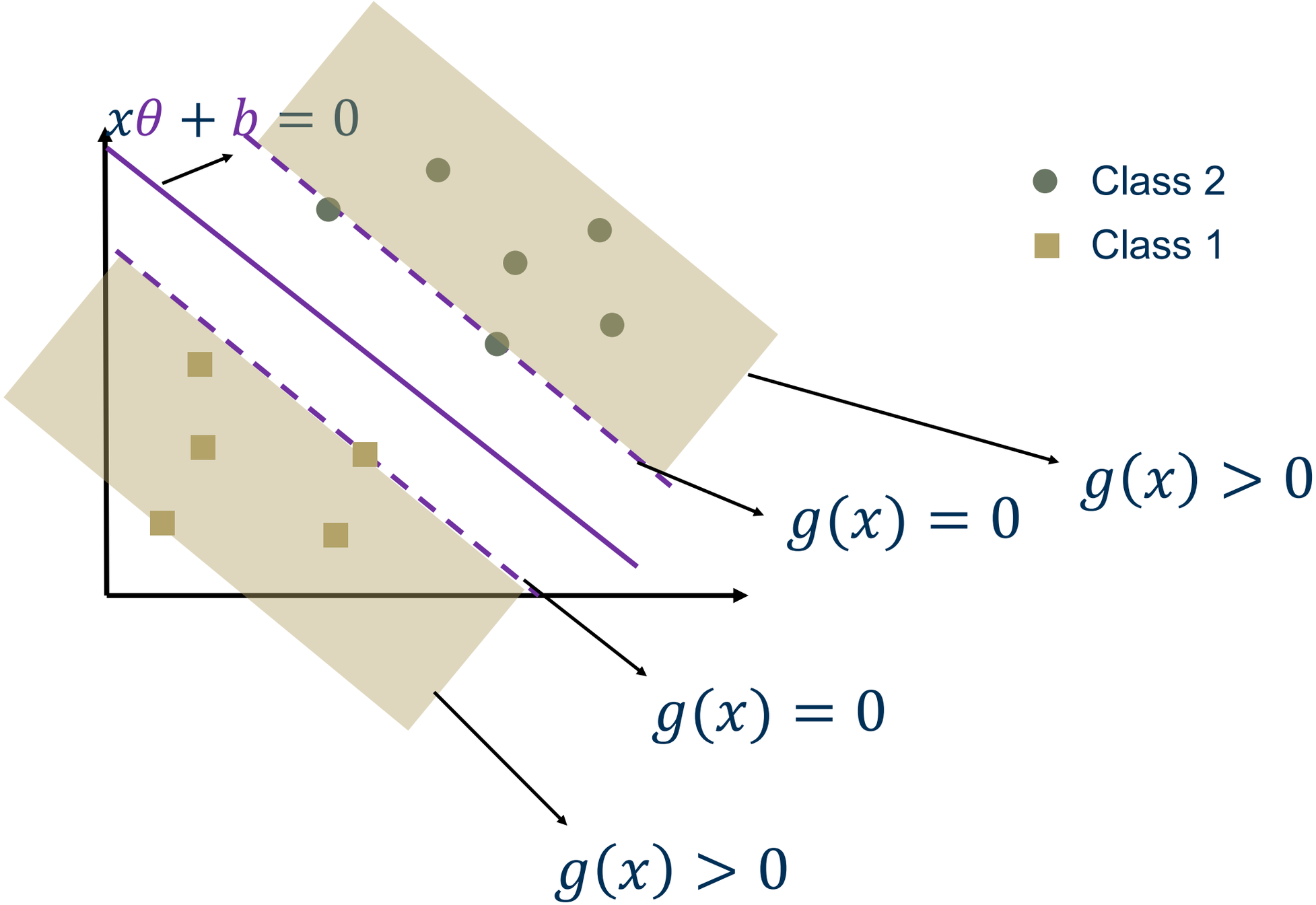

$$g(x) = y^{\{i\}}\left(x^{\{i\}}\theta + b\right) - 1$$

3) $g(x)\alpha = 0$ Complementary slackness

$$\Rightarrow \begin{cases} g(x) > 0, & \alpha = 0 \\ \alpha > 0, & g(x) = 0 \end{cases}$$

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Lagrange formulation (Primal → Dual Form)

Minimize $\frac{1}{2}\theta\theta^T$ s.t. $y^{\{i\}}\left(x^{\{i\}}\theta + b\right) - 1 \geq 0$

$$\mathcal{L}(\theta, b, \alpha) = \frac{1}{2}\theta\theta^T - \sum_{i=1}^{N} \alpha_i (y^{\{i\}}\left(x^{\{i\}}\theta + b\right) - 1)$$

Minimize w.r.t θ and b and maximize *w.r.t* each $\alpha_i \geq 0$

$$\nabla_\theta \mathcal{L}(\theta, b, \alpha) = \theta - \sum_{i=1}^{N} \alpha_i y^{\{i\}} x^{\{i\}} = 0$$

$$\nabla_b \mathcal{L}(\theta, b, \alpha) = -\sum_{i=1}^{N} \alpha_i y^{\{i\}} = 0$$

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$$\theta = \sum_{i=1}^{N} \alpha_i y^{\{i\}} x^{\{i\}}$$

$$\sum_{i=1}^{N} \alpha_i y^{\{i\}} = 0$$

Let's substitute these in the Lagrangian:

$$\mathcal{L}(\theta, b, \alpha) = \frac{1}{2}\theta\theta^T - \sum_{i=1}^{N} \alpha_i (y^{\{i\}}\left(x^{\{i\}}\theta + b\right) - 1)$$

$$\mathcal{L}(\theta, b, \alpha) = \sum_{i=1}^{N} \alpha_i + \frac{1}{2}\theta\theta^T - \sum_{i=1}^{N} \alpha_i (y^{\{i\}}\left(x^{\{i\}}\theta + b\right))$$

$$\mathcal{L}(\theta, b, \alpha) = \sum_{i=1}^{N} \alpha_i + \frac{1}{2}\theta\theta^T - \sum_{i=1}^{N} \alpha_i \left(y^{\{i\}}\left(x^{\{i\}}\theta\right)\right)$$

$$= \sum_{i=1}^{N} \alpha_i + \frac{1}{2}\theta\theta^T - \theta\theta^T = \sum_{i=1}^{N} \alpha_i - \frac{1}{2}\theta\theta^T$$

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$$\theta = \sum_{i=1}^{N} \alpha_i y^{\{i\}} x^{\{i\}}$$

$$\sum_{i=1}^{N} \alpha_i y^{\{i\}} = 0$$

$$\mathcal{L}(\theta, b, \alpha) = \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \theta \theta^T$$

$$\mathcal{L}(\theta, b, \alpha) = \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} y^{\{i\}} y^{\{j\}} \alpha_i \alpha_j x^{\{i\}} x^{\{j\}^T}$$

maximize *w.r.t* each $\alpha_i \geq 0$ for $i = 1, \ldots, N$

and

$$\sum_{i=1}^{N} \alpha_i y^{\{i\}} = 0$$

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The solution – quadratic programming

$$\max_{\alpha} \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} y^{\{i\}} y^{\{j\}} \alpha_i \alpha_j x^{\{i\}} x^{\{j\}^T}$$

Quadratic programming packages usually use “min”

$$\min_{\alpha} \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} y^{\{i\}} y^{\{j\}} \alpha_i \alpha_j x^{\{i\}} x^{\{j\}^T} - \sum_{i=1}^{N} \alpha_i$$

$$\min_{\alpha} \frac{1}{2} \alpha^T \begin{bmatrix} y^{\{1\}} y^{\{1\}} x^{\{1\}} x^{\{1\}^T} & \cdots & y^{\{1\}} y^{\{N\}} x^{\{1\}} x^{\{N\}^T} \\ \vdots & \ddots & \vdots \\ y^{\{N\}} y^{\{1\}} x^{\{N\}} x^{\{1\}^T} & \cdots & y^{\{N\}} y^{\{N\}} x^{\{N\}} x^{\{N\}^T} \end{bmatrix} \alpha + (-I^T) \alpha$$

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$$\min_{\alpha} \frac{1}{2} \alpha^T \underbrace{\begin{bmatrix} y^{\{1\}} y^{\{1\}} x^{\{1\}} x^{\{1\}^T} & \cdots & y^{\{1\}} y^{\{N\}} x^{\{1\}} x^{\{N\}^T} \\ \vdots & \ddots & \vdots \\ y^{\{N\}} y^{\{1\}} x^{\{N\}} x^{\{1\}^T} & \cdots & y^{\{N\}} y^{\{N\}} x^{\{N\}} x^{\{N\}^T} \end{bmatrix}}_{\text{Quadratic coefficients}} \alpha + \underbrace{(-I^T)\alpha}_{\text{Linear term}}$$

Subject to $\underbrace{\sum_{i=1}^{N} \alpha_i y^{\{i\}} = y^T \alpha = 0}_{\text{Linear equality constraint}}$

$$lower\ bound(0) \leq \quad \alpha \quad \leq \text{upper bound}(\infty)$$

Pass these to a quadratic programming package

$$\min_{\alpha} \frac{1}{2}\alpha^T Q\alpha - 1^T\alpha \quad \text{subject to} \quad y^T\alpha = 0; \alpha \geq 0$$

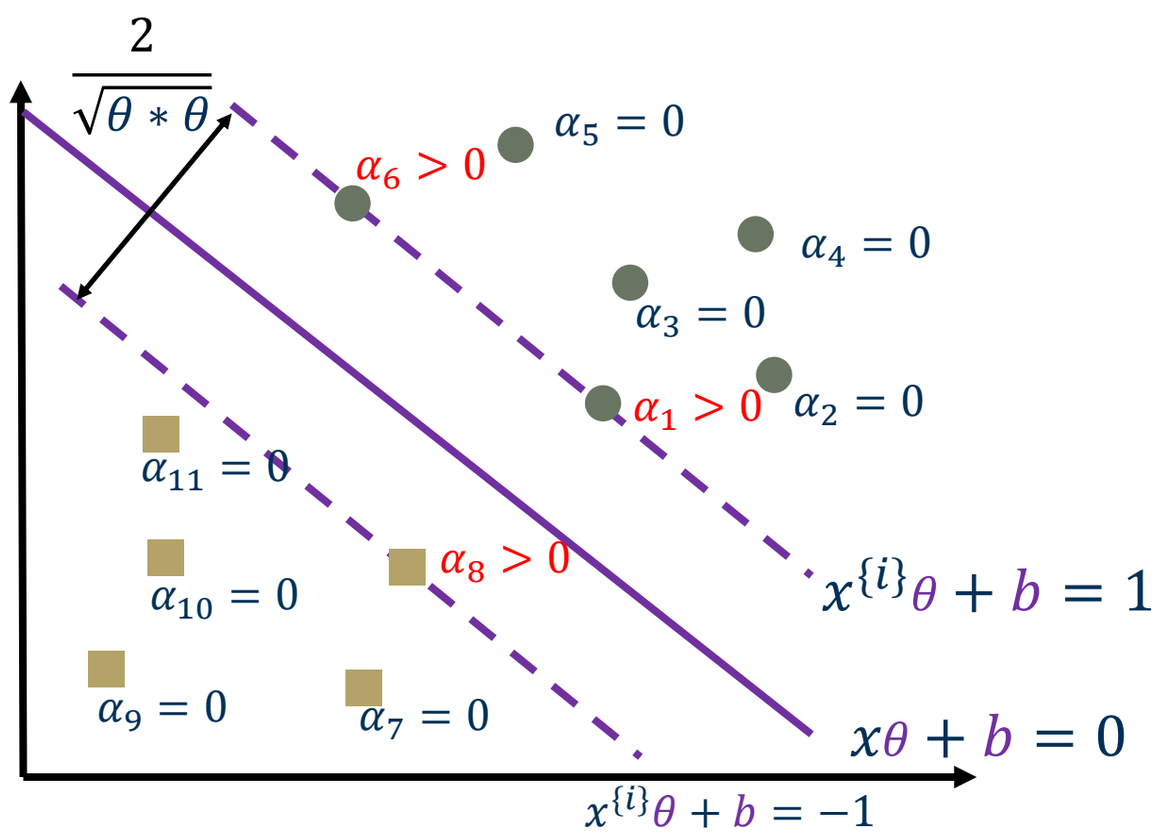
Quadratic programming will give us α

Solution: $\alpha = \alpha_1, \dots, \alpha_N$

KKT condition ($\alpha_i g_i(\theta) = 0$): $\alpha_i\left(y^{\{i\}}\left(x^{\{i\}}\theta + b\right) - 1\right) = 0$

$\left(y^{\{i\}}\left(x^{\{i\}}\theta + b\right) - 1\right) > 0 \quad \Rightarrow \quad \alpha_i = 0$

$\left(y^{\{i\}}\left(x^{\{i\}}\theta + b\right) - 1\right) = 0 \quad \Rightarrow \quad \alpha_i > 0 \Rightarrow x_i \; is \; a \; support \; vector$



Class 2

Class 1

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Training

$$\theta = \sum_{i=1}^{N} \alpha_i y^{\{i\}} x^{\{i\}}$$

No need to go over all datapoints

$$\rightarrow \theta = \sum_{x^{\{i\}} in\ SV} \alpha_i y^{\{i\}} x^{\{i\}}$$

and for b pick any support vector and calculate:

$$y^{\{i\}}\left(x^{\{i\}}\theta + b\right) = 1$$

Testing

For a new test point s

Compute:

$$s\theta + b = \sum_{x^{\{i\}} in\ SV} \alpha_i y^{\{i\}} x^{\{i\}} s^T + b$$

Classify s as class 1 if the result is positive, and class 2 otherwise

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Why primal → dual form?

Primal Form

Minimize $\frac{1}{2}\theta\theta^T$ s.t. $y^{\{i\}}\left(x^{\{i\}}\theta + b\right) - 1 \geq 0$

Dual Form

$$\min_{\alpha} \frac{1}{2}\alpha^T \begin{bmatrix} y^{\{1\}}y^{\{1\}}x^{\{1\}}x^{\{1\}^T} & \cdots & y^{\{1\}}y^{\{N\}}x^{\{1\}}x^{\{N\}^T} \\ \vdots & \ddots & \vdots \\ y^{\{N\}}y^{\{1\}}x^{\{N\}}x^{\{1\}^T} & \cdots & y^{\{N\}}y^{\{N\}}x^{\{N\}}x^{\{N\}^T} \end{bmatrix} \alpha + (-I^T)\alpha$$

s.t. $\sum_{i=1}^{N} \alpha_i y^{\{i\}} = y^T\alpha = 0$ s.t. $\alpha \geq 0$

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Why primal → dual form?

- **Constraints become easier**
  - Primal has **N inequality constraints**. Dual replaces constraints with $\alpha_i \geq 0$ and one equality constraint.
  - Optimization becomes a standard **Quadratic Program (QP)**.
- **Only support vectors matter**
  - Dual solution expresses $\theta = \sum_{i=1}^{N} \alpha_i \, y^{(i)} x^{(i)}$
  - Most $\alpha_i = 0 \rightarrow$ **training depends only on support vectors**, not all data.
- **Enables the kernel trick**
  - Dual objective depends only on **dot products**: $x^{(i)} \cdot x^{(j)}$
  - Replace dot product with kernel. $K\left(x^{(i)}, x^{(j)}\right) = \phi\left(x^{(i)}\right)^T \phi\left(x^{(j)}\right)$
  - Lets SVM work in **very high-dimensional (even infinite-dimensional)** spaces to solve the issue that comes with hard SVM
    - works for only linearly separable data
- **Strong duality holds**
- Dual optimum = primal optimum $\rightarrow$ solving the dual is **equivalent** and easier.

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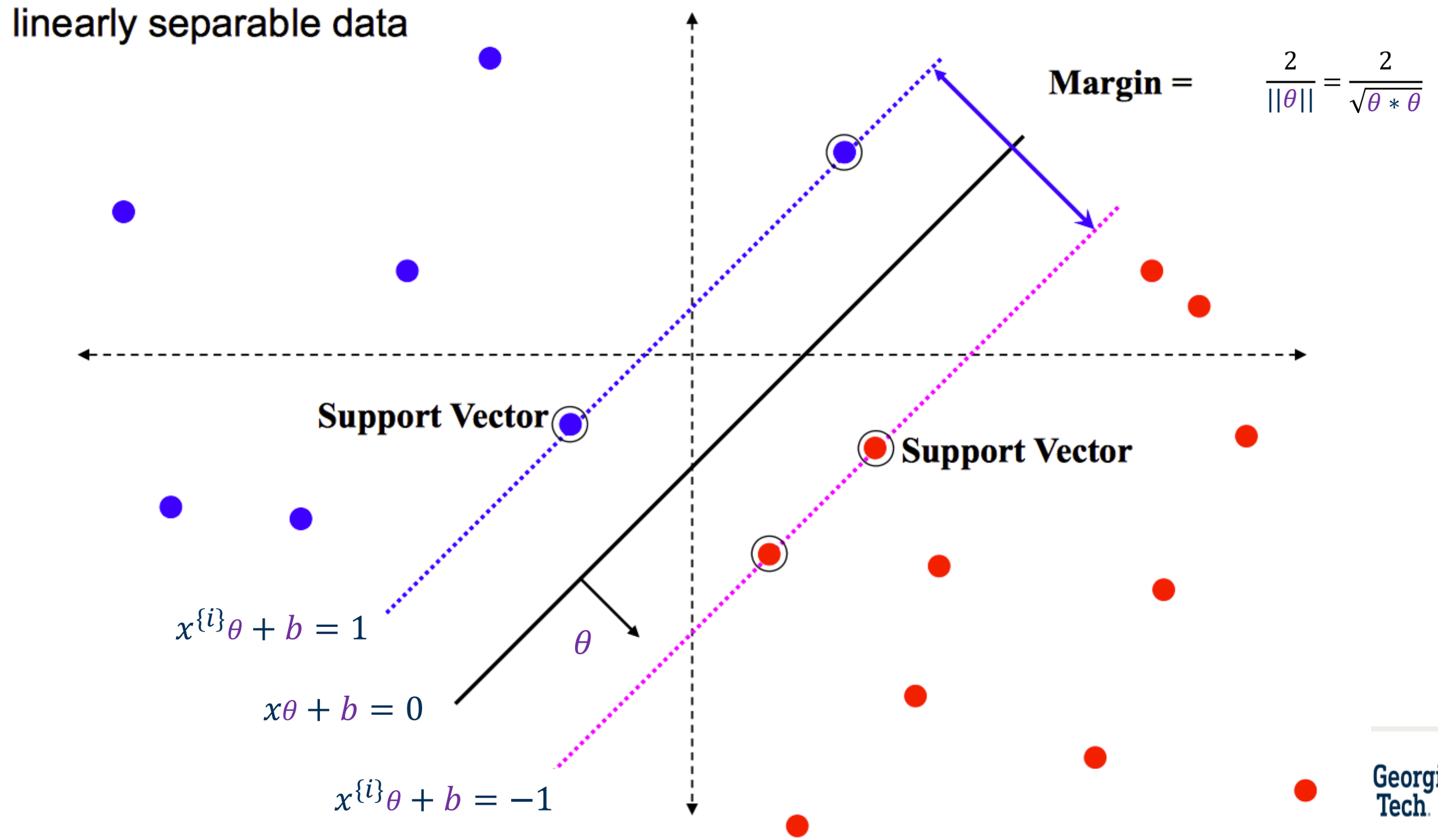
Geometric Interpretation



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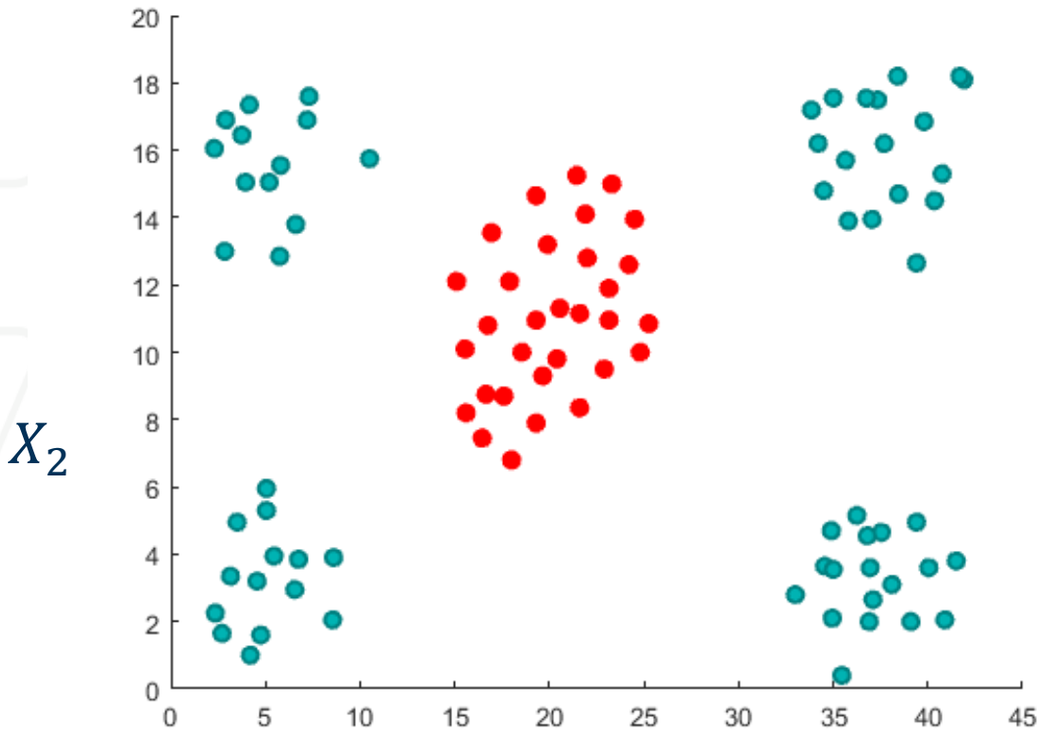
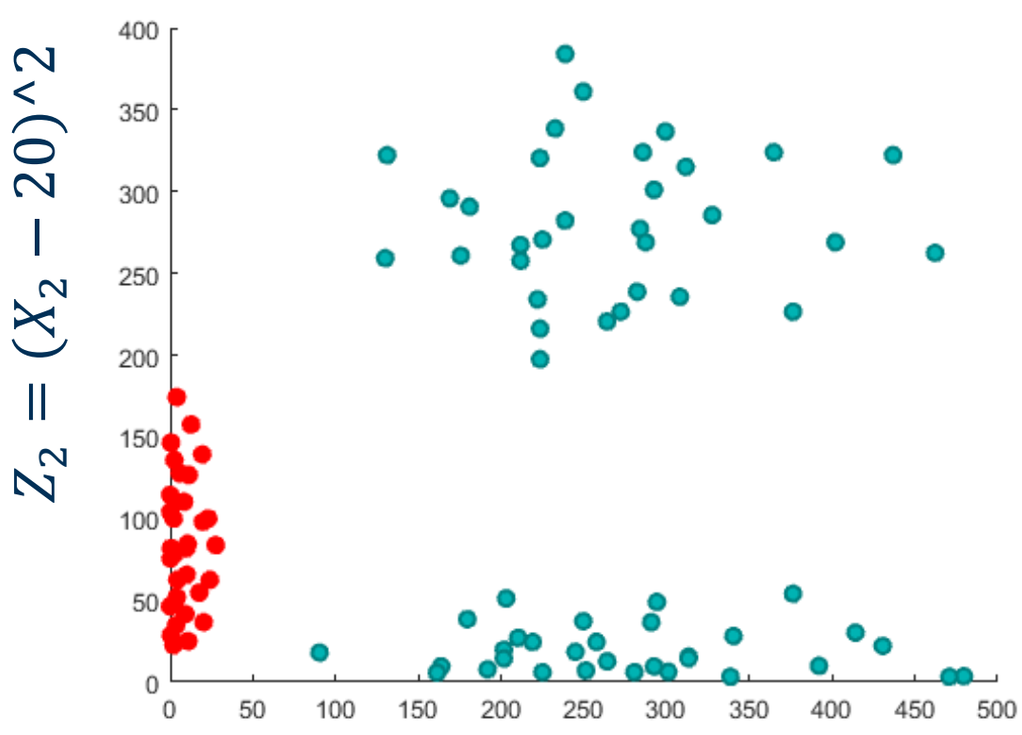
From x to z space





$X \longrightarrow Z$

(X_1^2, X_2^2)

In x space

$$\max_{\alpha} \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} y^{\{i\}} y^{\{j\}} \alpha_i \alpha_j x^{\{i\}} x^{\{j\}^T}$$

$$X_2 \max_{\alpha} \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} y^{\{i\}} y^{\{j\}} \alpha_i \alpha_j x^{\{i\}} x^{\{j\}^T}$$
$$X_1$$

$let's\ say\ x\ is\ n \times d$

xx^T will be $n \times n$

If I add millions of dimensions to x, would it affect the final size of xx^T?

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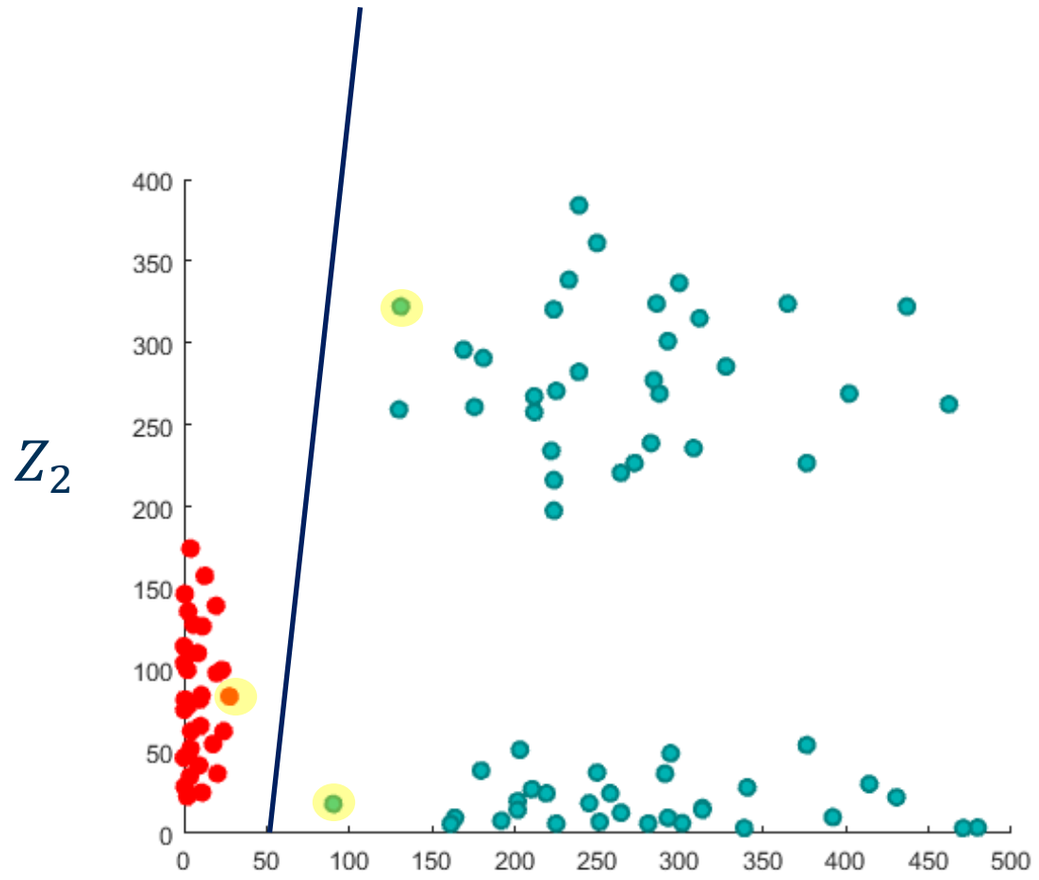
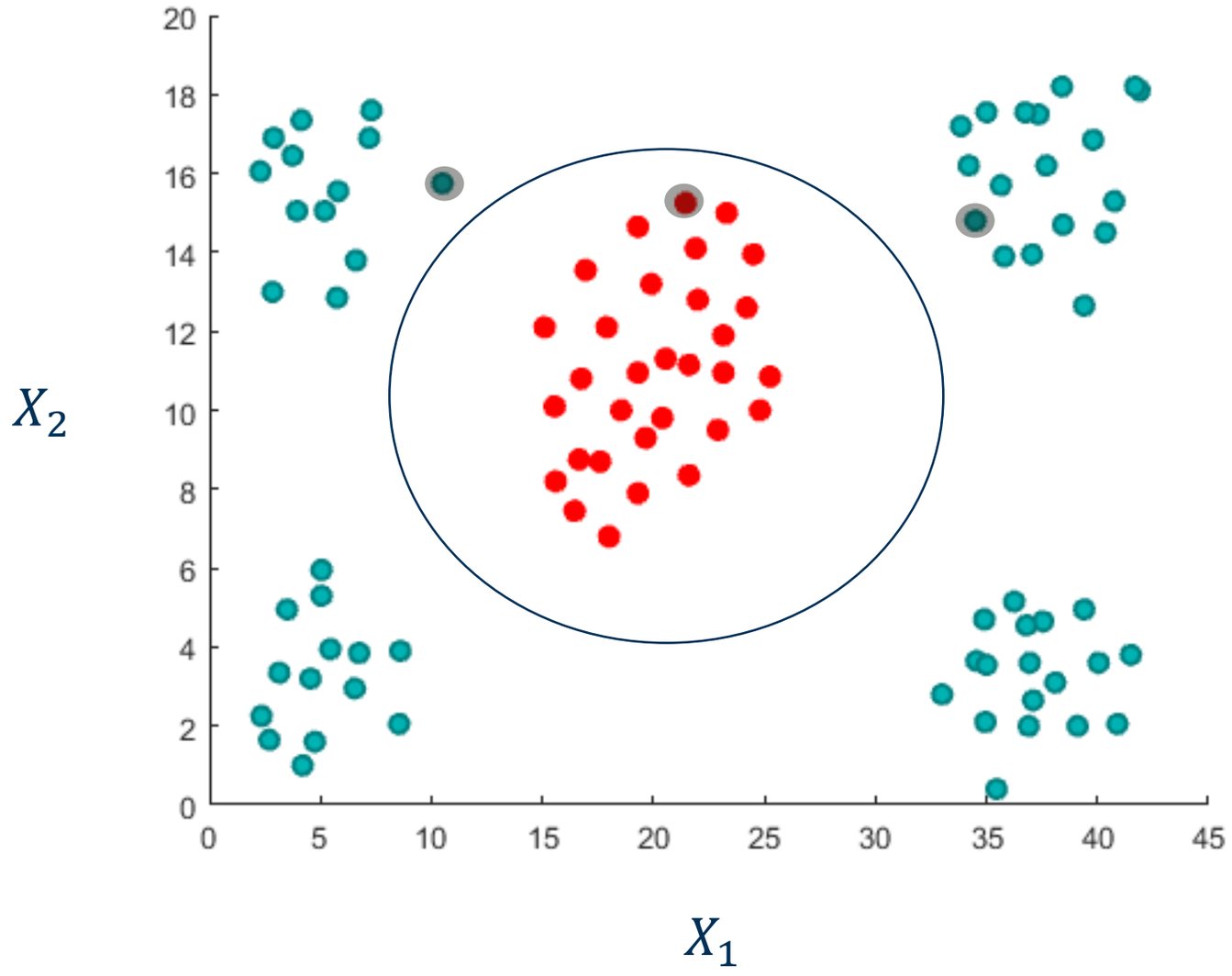
In z space

$$\max_{\alpha} \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} y^{\{i\}} y^{\{j\}} \alpha_i \alpha_j z^{\{i\}} z^{\{j\}^T}$$



In x space, they are called pre-images of support vectors



Take-Home Messages

- Linear Separability
- Perceptron
- SVM: Geometric Intuition and Formulation

Quick Knowledge Check

- What is the main motivation for SVM compared to perceptron? A. Removes bias from the classifier B. Finds the separating hyperplane with the largest margin C. Forces all margins to be equal D. Ensures nonlinear boundaries
- To enforce correct classification with margin, SVM requires: A. $y^{\{i\}}(x^{\{i\}}\theta + b) \leq 0$ B. $x^{\{i\}}\theta + b \geq 0$ C. $y^{\{i\}}\theta = 1$ D. $y^{\{i\}}(x^{\{i\}}\theta + b) \geq 1$
- The primal SVM objective is: A. Maximize $\theta^T\theta$ B. Maximize the number of support vectors C. Minimize $(1/2)\theta^T\theta$ D. Minimize $x^{\{i\}}\theta$
- Why do we convert the primal SVM to the dual? A. To avoid support vectors B. To increase dimensionality C. To allow kernels and handle constraints more easily D. To remove θ entirely
- In testing with SVM, the classification of a point s is based on: A. $\Sigma\alpha_i$ only B. b only C. distance to nearest sample D. $\text{sign}(\theta^T s + b)$
- When is a training point a support vector? A. When $y^{\{i\}}(x^{\{i\}}\theta + b) > 2$ B. When $\alpha_i = 0$ C. When $y^{\{i\}}(x^{\{i\}}\theta + b) = 1$ D. When $x^{\{i\}}$ lies inside the margin

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